

# UNIVERSITY OF SCIENCE AND TECHNOLOGY OF SOUTHERN PHILIPPINES

## CS 414 - 4B - ACTIVITY 1 (Part 1): Havel-Hakimi Algorithm

### Group Members:

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### Problem No. 5

A social network analyst wants to map interactions among 6 club members with the target degree sequence:  $S_5 = (5, 4, 3, 2, 1, 3)$ .

#### a. Determining Whether the Sequence is Graphical

##### Preliminary Check:

Number of vertices ( $n$ ) = 6

Sum of degrees =  $5 + 4 + 3 + 2 + 1 + 3 = 18$  (even, necessary condition satisfied)

Maximum degree =  $5 = n - 1$  (does not exceed the maximum possible degree)

$S_5$  is given unsorted. It is sorted to  $(5, 4, 3, 3, 2, 1)$  before applying the algorithm, since Havel-Hakimi requires non-increasing order at every step. Sorting reorders the working list only — each vertex keeps the degree assigned to it by the given sequence.

#### Applying the Havel-Hakimi Algorithm:

##### Step 1:

Sort in non-increasing order:  $(5, 4, 3, 3, 2, 1)$

Remove the first element (5).

Subtract 1 from the next 5 elements:

$(4-1, 3-1, 3-1, 2-1, 1-1) = (3, 2, 2, 1, 0)$

No negative values. Proceed.

##### Step 2:

Sort in non-increasing order:  $(3, 2, 2, 1, 0)$

Remove the first element (3).

Subtract 1 from the next 3 elements:

$(2-1, 2-1, 1-1, 0) = (1, 1, 0, 0)$

No negative values. Proceed.

##### Step 3:

Sort in non-increasing order:  $(1, 1, 0, 0)$

Remove the first element (1).

Subtract 1 from the next 1 element:

$$(1-1, 0, 0) = (0, 0, 0)$$

No negative values. Proceed.

Step 4:

Remaining sequence: (0, 0, 0)

All elements are zero.

**Conclusion:** The sequence  $S_5 = (5, 4, 3, 2, 1, 3)$  IS GRAPHICAL. A simple graph with this exact degree sequence can be constructed.

## b. Graph Construction

**Vertex Assignment** (by position in the given sequence  $S_5 = (5, 4, 3, 2, 1, 3)$ ):

$v_1 = \text{degree } 5$        $v_3 = \text{degree } 3$        $v_5 = \text{degree } 1$

$v_2 = \text{degree } 4$        $v_4 = \text{degree } 2$        $v_6 = \text{degree } 3$

**Working order for the algorithm** (sorted by degree, labels unchanged):

$v_1(5), v_2(4), v_3(3), v_6(3), v_4(2), v_5(1)$

**Edge List** (derived by tracing the algorithm):

From Step 1 ( $v_1$  connects to all other vertices):

$v_1-v_2, v_1-v_3, v_1-v_6, v_1-v_4, v_1-v_5$

From Step 2 ( $v_2$  connects to the next 3 highest-degree vertices):

$v_2-v_3, v_2-v_6, v_2-v_4$

From Step 3 ( $v_3$  connects to the next 1 highest-degree vertex):

$v_3-v_6$

Total edges: 9

## Adjacency Matrix:

|       | $v_1$ | $v_2$ | $v_3$ | $v_4$ | $v_5$ | $v_6$ |
|-------|-------|-------|-------|-------|-------|-------|
| $v_1$ | [ 0   | 1     | 1     | 1     | 1     | 1 ]   |
| $v_2$ | [ 1   | 0     | 1     | 1     | 0     | 1 ]   |
| $v_3$ | [ 1   | 1     | 0     | 0     | 0     | 1 ]   |
| $v_4$ | [ 1   | 1     | 0     | 0     | 0     | 0 ]   |
| $v_5$ | [ 1   | 0     | 0     | 0     | 0     | 0 ]   |
| $v_6$ | [ 1   | 1     | 1     | 0     | 0     | 0 ]   |

## Verification:

$v_1$ : connected to  $v_2, v_3, v_4, v_5, v_6$       -> degree 5      OK

$v_2$ : connected to  $v_1, v_3, v_4, v_6$       -> degree 4      OK

$v_3$ : connected to  $v_1, v_2, v_6$       -> degree 3      OK

$v_4$ : connected to  $v_1, v_2$       -> degree 2      OK

$v_5$ : connected to  $v_1$       -> degree 1      OK

$v_6$ : connected to  $v_1, v_2, v_3$       -> degree 3      OK

Sum of degrees =  $5 + 4 + 3 + 2 + 1 + 3 = 18 = 2 \times 9$  edges

The graph satisfies the given degree sequence  $S_5 = (5, 4, 3, 2, 1, 3)$ .

**Graph Drawing:**

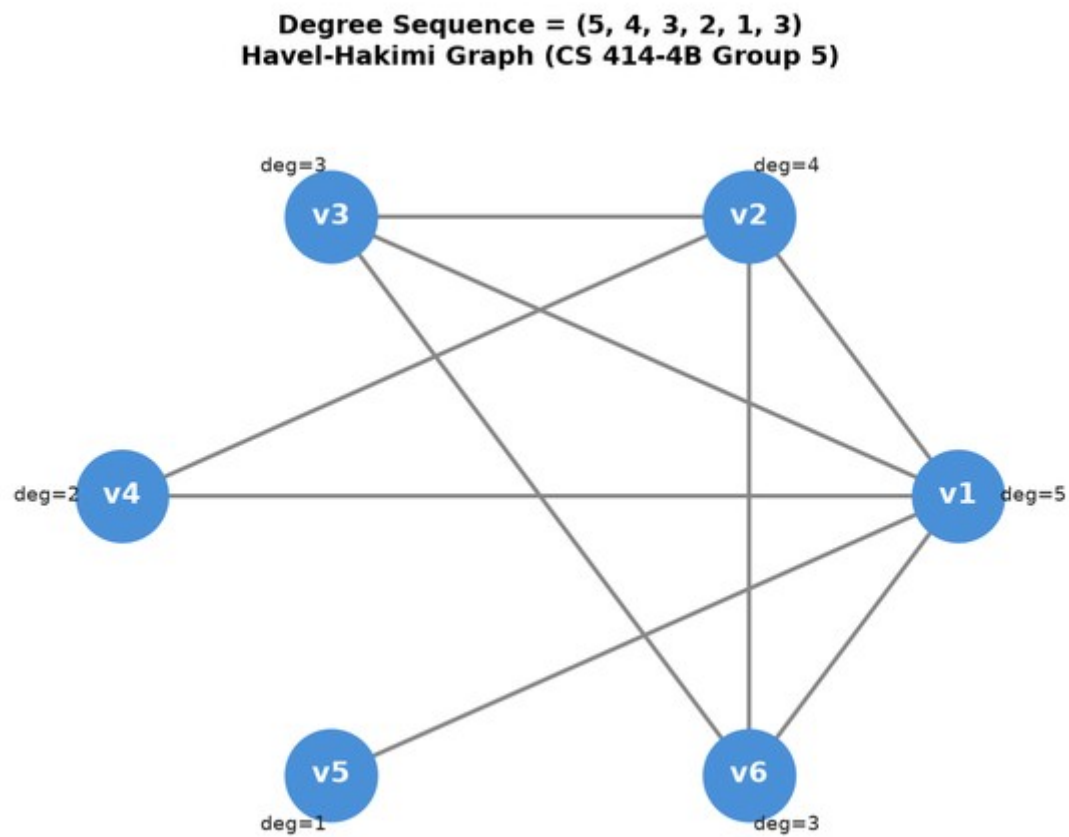


Figure 1. Constructed graph for  $S_5 = (5, 4, 3, 2, 1, 3)$ .  
Each vertex is labeled with its identifier and its degree.